MS&E 226 — Midterm 1 Practice Questions (Generated Set 1)

10 questions, deliberately weighted toward Lectures 7–9 (bias-variance decomposition, frequentist inference, standard errors & confidence intervals for the sample mean) — the material with the least existing practice-question coverage relative to how much class time it received. Multiple choice, one correct answer among (a)–(d); "(e)" lets you skip a question, matching last year's exam format.

Question 1

A learning algorithm predicts an outcome at a fixed covariate vector (x). At this point, the irreducible error is $\sigma^2((x)) = 9$, the bias is -2, and the variance is 5. What is the resulting Average MSE?

(a) 18

(b) 12

(c) 9

(d) $\sqrt{18} \approx 4.24$

(e) I choose not to answer this question.

Question 2

A fellow student claims: "There's always a strict bias-variance tradeoff: whenever you change your learning algorithm so that variance decreases, bias must increase, and vice versa." Which of the following best evaluates this claim, based on what was covered in lecture?

(a) Correct — this follows directly from the bias-variance decomposition, since Average MSE = irreducible error + Bias^2 + Variance, and the three terms must always sum to a fixed total.

(b) Correct — the Gauss-Markov theorem shows OLS has the lowest variance among all zero-bias estimators using a given feature set, so no learning algorithm can ever have both lower bias and lower variance than another.

(c) Incorrect — the decomposition is just that, a decomposition, not a conservation law. Nothing forces bias and variance to move in opposite directions, and lecture gave explicit examples (double descent; adding genuinely informative features) where both bias and variance decreased at the same time.

(d) Incorrect — in fact the opposite is true: bias and variance always move together in the **same** direction as model complexity increases.

(e) I choose not to answer this question.

Question 3

Assume (A1)–(A3) hold (linear population model; all relevant features observed; homoskedastic, mean-zero errors). A data scientist fits a ridge regression model with $\lambda > 0$ on the same features used by an

OLS model, and finds the ridge model has strictly lower variance than OLS at every test point. What must be true of the ridge model's bias, and why?

(a) It must be exactly zero too, since ridge regression — like OLS — is a linear, unbiased estimator under (A1)–(A3).

(b) It must be nonzero (strictly positive, in the sense of having nonzero bias at some point), because the Gauss-Markov theorem says OLS has the lowest variance among **all** linear, unbiased estimators using these features — so a linear estimator with strictly lower variance than OLS cannot also be unbiased.

(c) It could be anything — positive, negative, or zero — since bias and variance are unrelated quantities under (A1)–(A3).

(d) It must be **lower** than OLS's bias (i.e., closer to zero), since regularization is specifically designed to correct bias in the OLS estimate.

(e) I choose not to answer this question.

Question 4

Assume (A1)–(A3) hold for a population model with three relevant features A, B, C: $Y = \beta_A A + \beta_B B + \beta_C C + \varepsilon$. A data scientist fits OLS using only A and B, omitting C. Which of the following is the most accurate statement about the resulting model?

(a) The model is unbiased — OLS remains BLUE even when a relevant feature is omitted, as long as the remaining features are observed correctly.

(b) The model is generally biased, and the size of that bias does not depend at all on how C is correlated with A and B — omitting any relevant feature produces the same bias regardless of correlation structure.

(c) The model is generally biased, and the size of the bias — along with any distortion to the fitted coefficients on A and B themselves — depends on how correlated the omitted feature C is with A and B.

(d) The model is biased, but only in its predictions — the fitted coefficients on A and B remain correct (unbiased) estimates of β_A and β_B.

(e) I choose not to answer this question.

Question 5

In Lecture 7, the bias-variance decomposition for a prediction $\hat{f}((x))$ includes an irreducible error term $\sigma^2((x))$. In Lecture 8, the analogous decomposition for an estimator $\hat{\theta}$ of a population parameter θ is simply $\text{MSE} = \text{Bias}^2 + \text{Variance}$, with **no** irreducible error term. Why does the estimator decomposition have no irreducible error term?

(a) Because estimators are always unbiased, so there is nothing left to attribute to irreducible error.

(b) Because the irreducible error term only matters for classification problems, not for estimating a continuous parameter.

(c) Because the true outcome Y being predicted in the prediction problem is itself random even conditional on (X), whereas the parameter θ being estimated is treated as a **fixed** (non-random) quantity under the frequentist view — so there's no analogous "irreducible" randomness in the target itself.

(d) Because irreducible error only arises when the sample size is small; for the large-n asymptotic theory used in estimation, it vanishes.

(e) I choose not to answer this question.

Question 6

Under the frequentist view of statistical inference, which of the following is randomly varying across "parallel universes," and which is fixed?

(a) The population parameter θ is random; the data sample is fixed.

(b) Both the population parameter θ and the data sample are random.

(c) The population parameter θ is fixed; the data sample (and therefore any estimator computed from it) is random.

(d) Both the population parameter θ and the data sample are fixed once the sample has been collected — there is no remaining randomness to reason about.

(e) I choose not to answer this question.

Question 7

You estimate a population mean μ using the sample mean of n i.i.d. draws. With a sample of size $n = 25$ you compute standard error $\widehat{\mathrm{SE}}_{25}$; with a much larger sample of size $n = 400$ (and roughly the same sample standard deviation $\hat{\sigma}$) you compute $\widehat{\mathrm{SE}}_{400}$. Approximately how do these compare?

(a) $\widehat{\mathrm{SE}}_{25}$ is about 4 times larger than $\widehat{\mathrm{SE}}_{400}$.

(b) $\widehat{\mathrm{SE}}_{25}$ is about 16 times larger than $\widehat{\mathrm{SE}}_{400}$.

(c) $\widehat{\mathrm{SE}}_{25}$ and $\widehat{\mathrm{SE}}_{400}$ are approximately equal, since standard error doesn't depend on sample size once $\hat{\sigma}$ is fixed.

(d) $\widehat{\mathrm{SE}}_{400}$ is about 4 times larger than $\widehat{\mathrm{SE}}_{25}$.

(e) I choose not to answer this question.

Question 8

After collecting a sample and computing a 95% confidence interval $[28.1, 35.7]$ for a population mean μ, a colleague says: "There is a 95% probability that μ is between 28.1 and 35.7." Is this an accurate interpretation, as discussed in lecture?

(a) Yes — by definition, a 95% confidence interval means there's a 95% probability the true parameter falls inside this specific interval.

(b) No — the 95% figure describes the **procedure**: across repeated samples, 95% of the resulting intervals would contain the true, fixed μ. Once a specific interval like $[28.1, 35.7]$ has been computed, μ either is or isn't in it; μ is not a random variable, so it doesn't make sense to assign it a probability.

(c) No — the correct statement is that there's a 95% probability the **sample mean** $\overline{Y}$ falls between 28.1 and 35.7.

(d) Yes, but only if the underlying population distribution is itself normal; otherwise the 95% figure doesn't apply.

(e) I choose not to answer this question.

Question 9

Holding the sample size n and sample standard deviation $\hat{\sigma}$ fixed, consider two confidence intervals for the population mean constructed from the same sample: a 95% CI and a 99% CI. Evaluate:

(i) The 99% CI is wider than the 95% CI.

(ii) The 99% CI is guaranteed to contain the true population mean μ, while the 95% CI is not.

Which of the following is correct?

(a) Only statement (i) is correct.

(b) Only statement (ii) is correct.

(c) Both statements are correct.

(d) Neither statement is correct.

(e) I choose not to answer this question.

Question 10

Consider the sample mean $\overline{Y}$ computed from n i.i.d. draws from a population with mean μ and variance σ^2. The same quantity $\overline{Y}$ can be analyzed two ways covered in this course: (I) as a **prediction** of a new, as-yet-unseen outcome Y from the same population, and (II) as an **estimator** of the population parameter μ itself. Which of the following correctly compares the Bias and Variance terms, and the overall MSE, across these two analyses?

(a) Bias and Variance are different in the two analyses, because predicting Y requires a different formula for $\overline{Y}$ than estimating μ.

(b) Bias and Variance are identical in the two analyses — in both cases Bias $= 0$ and Variance $= \frac{\sigma^2}{n}$ — but the overall MSE differs, because the prediction-MSE additionally includes the irreducible error term σ^2, which has no counterpart when the target is the fixed parameter μ.

(c) Bias and Variance are identical, and so is the overall MSE in both analyses, since $\overline{Y}$ is being compared to the same kind of target (μ) in both cases.

(d) Only the Variance term is identical across the two analyses ($\frac{\sigma^2}{n}$ in both cases); the Bias differs, because as a predictor $\overline{Y}$ is biased (due to the irreducible error), while as an estimator it is exactly unbiased.

(e) I choose not to answer this question.